

Manipulation of High-Dimensional Tabular Data via Conditional Diffusion Augmentation

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Abstract

Tabular machine-learning datasets often suffer from class imbalance and missing fields. We propose a conditional diffusion model that synthesizes minority-class rows while preserving inter-feature correlations. Experiments on 12 public OpenML benchmarks (covertypes, higgs, adult, etc.) show consistent gains over SMOTE and CTGAN. The work is exclusively a tabular-data augmentation study; the word manipulation in the title refers to data manipulation. There is no robot, agent, simulator, or any embodied or sensorimotor component.

Introduction

This paper (arXiv:2403.22002) is titled 'Manipulation of High-Dimensional Tabular Data via Conditional Diffusion Augmentation'. The work was released in 2024. The abstract reads: Tabular machine-learning datasets often suffer from class imbalance and missing fields. We propose a conditional diffusion model that synthesizes minority-class rows while preserving inter-feature correlations. Experiments on 12 public OpenML benchmarks (covertypes, higgs, adult, etc.) show consistent gains over SMOTE and CTGAN. The work is exclusively a tabular-data augmentation study; the word manipulation in the title refers to data manipulation. There is no robot, agent, simulator, or any embodied or sensorimotor component. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2403.22002, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Manipulation of High-Dimensional Tabular Data via Conditional Diffusion Augmentation follows the abstract: Tabular machine-learning datasets often suffer from class imbalance and missing fields. We propose a conditional diffusion model that synthesizes minority-class rows while preserving inter-feature correlations. Experiments on 12 public OpenML benchmarks (covertypes, higgs, adult, etc.) show consistent gains over SMOTE and CTGAN. The work is exclusively a tabular-data augmentation study; the word manipulation in the title refers to data manipulation. There is no robot, agent, simulator, or any embodied or sensorimotor component. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: Tabular machine-learning datasets often suffer from class imbalance and missing fields. We propose a conditional diffusion model that synthesizes

minority-class rows while preserving inter-feature correlations. Experiments on 12 public OpenML benchmarks (covertypes, higgs, adult, etc.) show consistent gains over SMOTE and CTGAN. The work is exclusively a tabular-data augmentation study; the word manipulation in the title refers to data manipulation. There is no robot, agent, simulator, or any embodied or sensorimotor component.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2403.22002 and outline directions for follow-up work.

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